

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating only

\$59.76

Avg Purchase

Demographics

Age, Gender, Location, Subscription Status

Behavior

Discount, Promo Code, Frequency, Review Rating, Shipping

Data Preparation in Python

01

Load & Explore

pandas · df.info() · describe()

02

Clean & Standardize

Imputed Review Rating (median by category) · snake_case columns

03

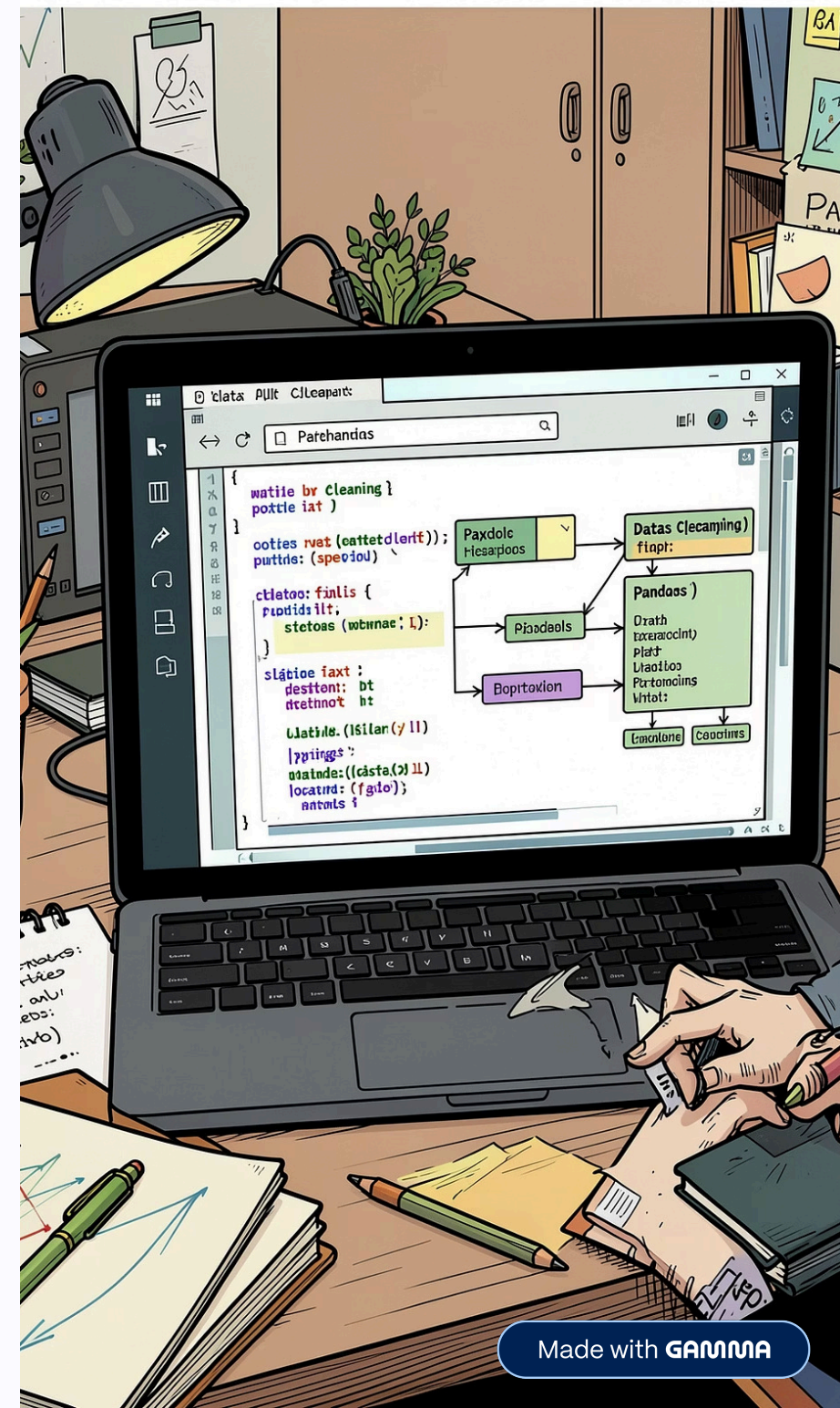
Feature Engineer

age_group bins · purchase_frequency_days

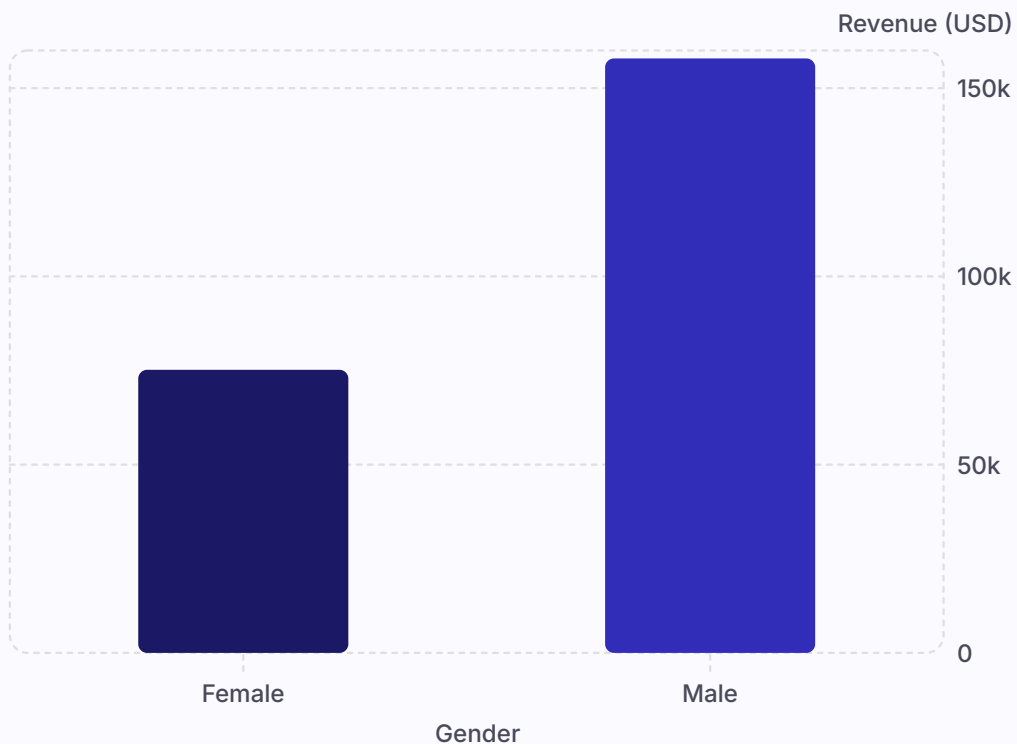
04

Load to PostgreSQL

Cleaned DataFrame → SQL



Revenue by Gender

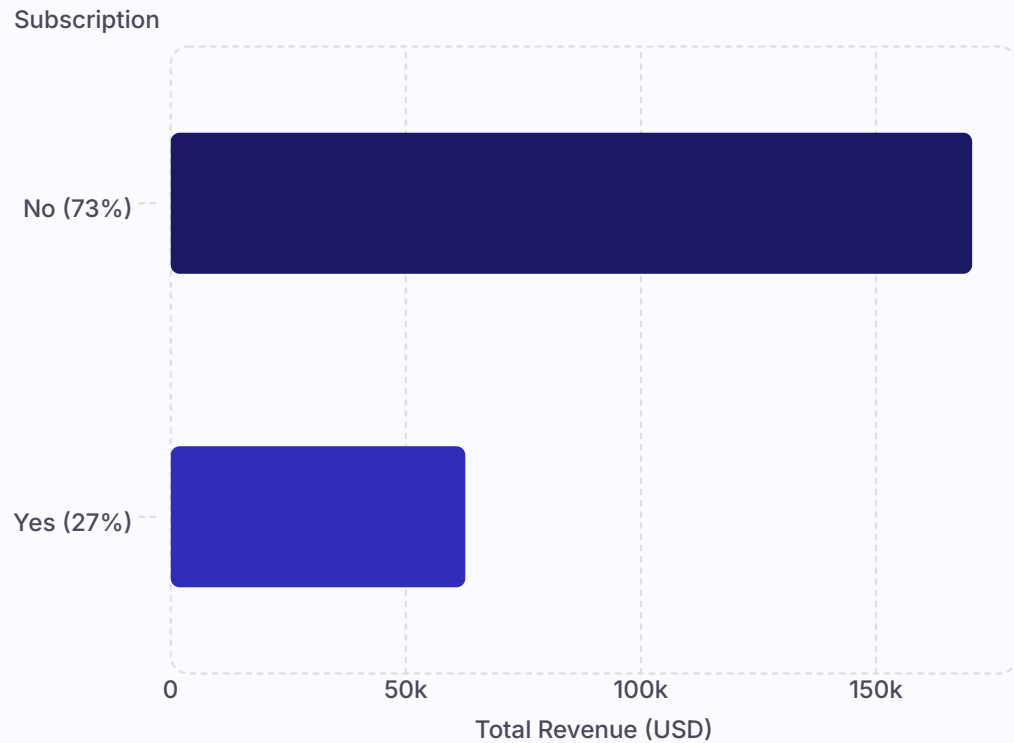


Key Insight

Male customers generated **2× more revenue** than female customers — \$157,890 vs \$75,191.

- ❏ Male shoppers dominate total spend despite representing fewer unique customers.

Subscribers vs Non-Subscribers



Subscription Breakdown

Non-Subscribers

2,847 customers · \$170K
revenue

Subscribers

1,053 customers · \$62K
revenue

Top Products & Categories

Top Rated

Gloves · Sandals · Boots · Hat · Skirt

Most Discounted

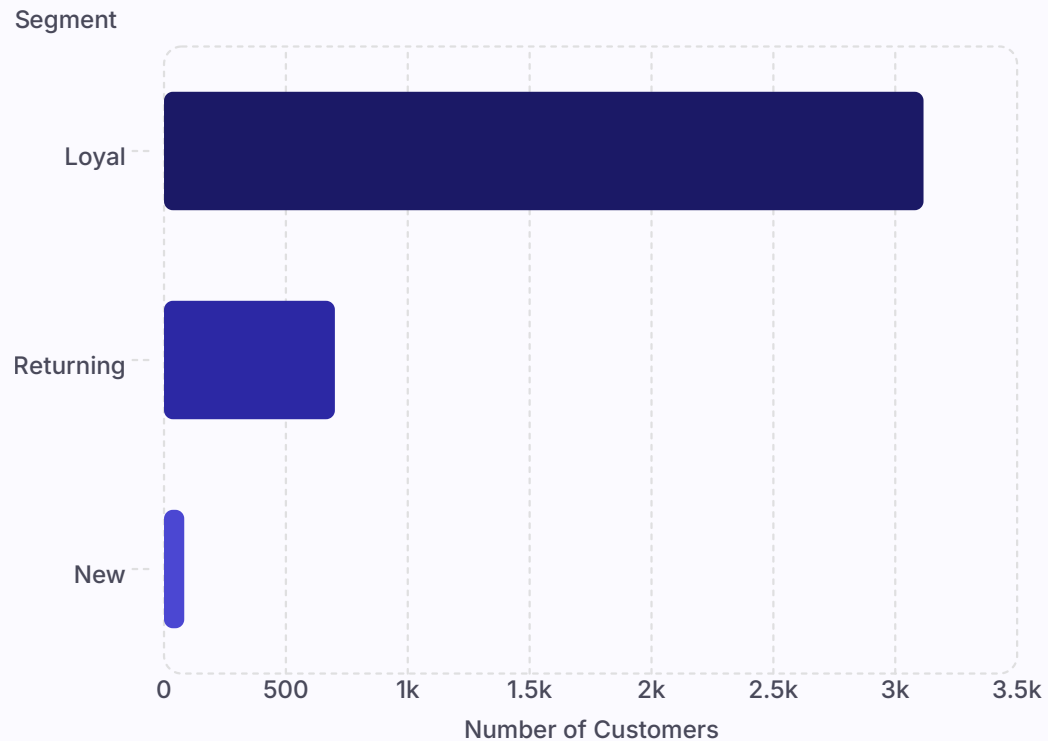
Hat (50%) · Sneakers · Coat · Sweater · Pants

Revenue by Category

Clothing ~\$100K · Accessories ~\$60K · Footwear ~\$30K · Outerwear ~\$15K



Customer Segmentation



Segment Definitions

Loyal — 3,116

High purchase frequency

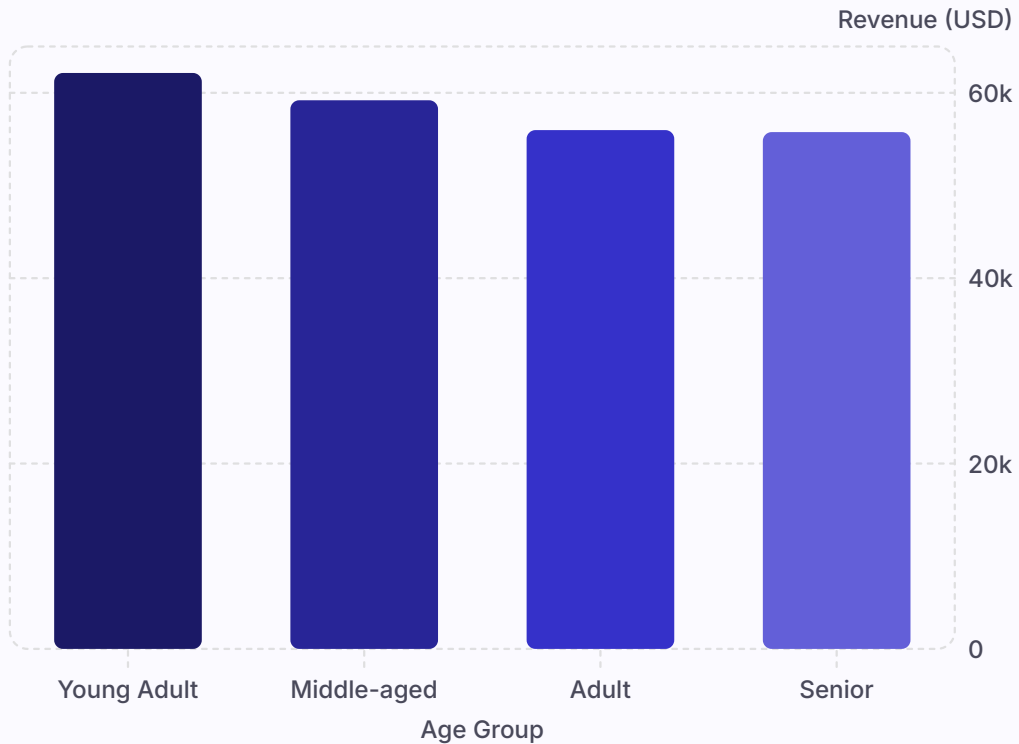
Returning — 701

Moderate repeat activity

New — 83

First-time buyers

Revenue by Age Group



Age Group Insights

Young Adults lead revenue at **\$62K**. Revenue declines gradually with age — all groups within ~\$6K of each other.

i Express shipping users spend slightly more on average (\$60.48 vs \$58.46).



Power BI Dashboard

3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Filters: Subscription · Gender · Category · Shipping Type

Business Recommendations



Boost Subscriptions

Exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers



Review Discounts

Balance sales vs. margins



Targeted Marketing

Focus on high-revenue age groups & express shippers